

The Problem

Potholes are still discovered too slowly: scheduled inspections, scattered citizen complaints and delayed municipal response leave many hazards untreated.

- Road damage creates safety risk and expensive vehicle repairs.
- Manual inspections miss many streets between scheduled checks.
- Reports must become assigned, visible repair tasks, not just map markers.

Our Solution

RoadSenseAI turns everyday driving recordings into a live, evidence-based pothole map and a managed repair workflow for city teams.

Citizen
uploads drive evidence

AI Pipeline
detects and merges
potholes

Control Room
oversees areas and
teams

Authority Team
assigns and repairs
work

Core idea: every detection becomes a traceable maintenance item until it is resolved.

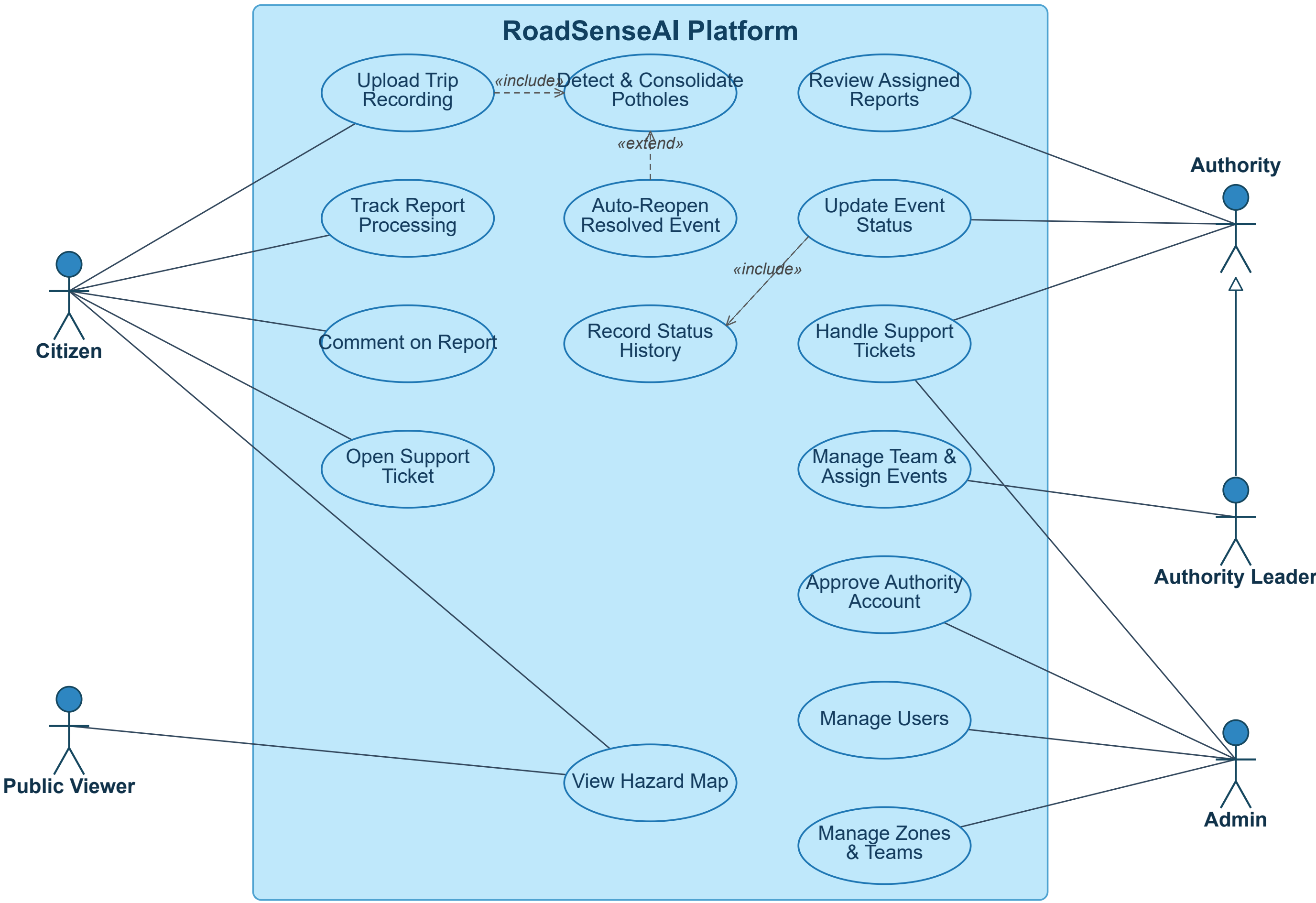
How It Works - One Pothole, One Managed Repair Story

CITIZEN AI SYSTEM CITY CONTROL AUTHORITY

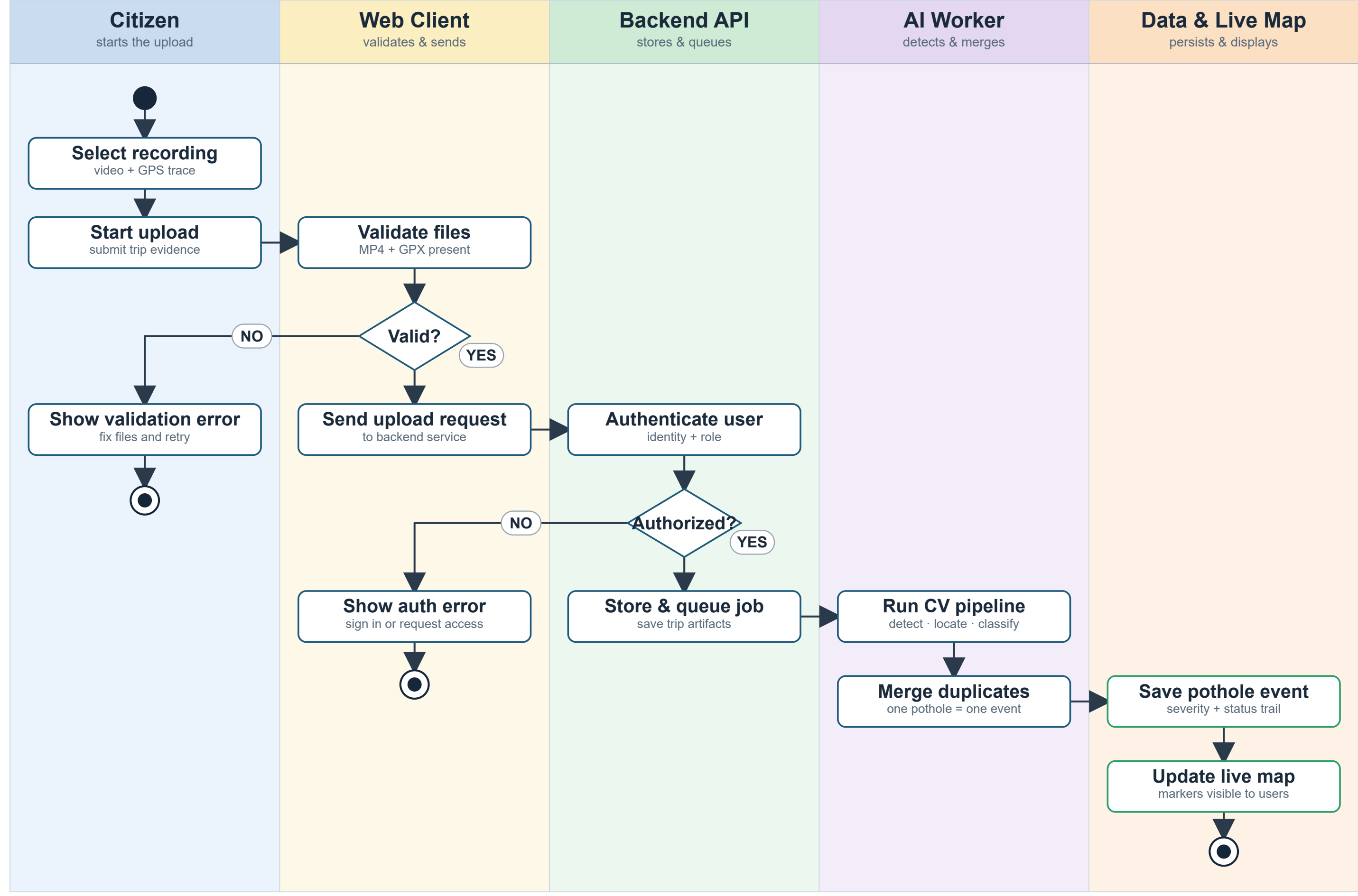


System Architecture - Use Case & Activity Diagrams

Use Case Diagram — RoadSenseAI Platform



Activity Diagram — Upload to Map



Inside the Detection Pipeline - 8 Stage UML Flow

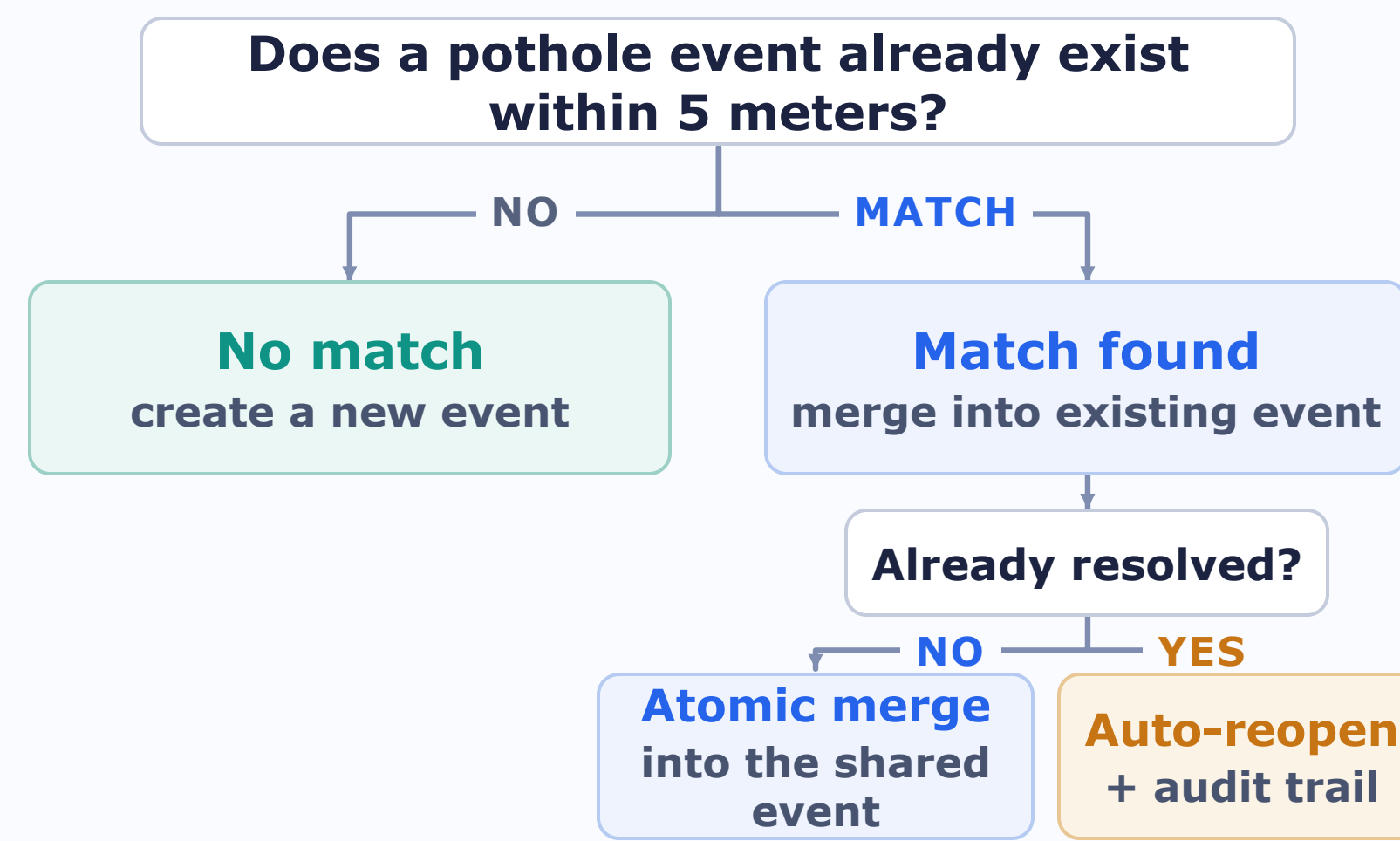
Worker pipeline: from trip artifacts to one managed pothole event



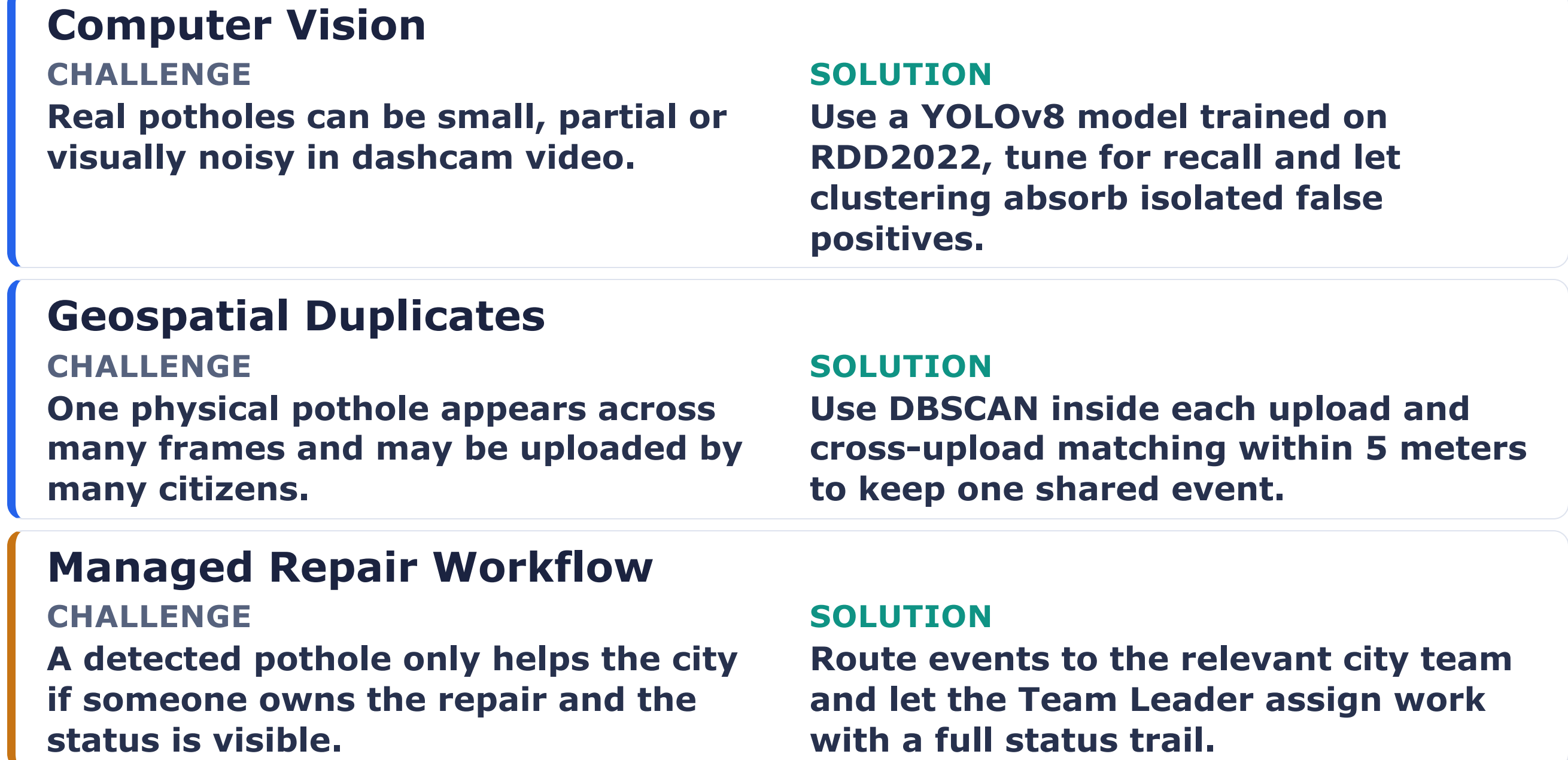
Key Tech React + Vite Leaflet FastAPI YOLOv8 OpenCV DBSCAN MongoDB Redis/RQ Docker

ALL STAGES RUN IN THE AI WORKER

Stage 8 decision



Engineering Challenges & Solutions



Core Requirements & Project KPIs

Functional

- Upload video + GPS and track processing
- Detect, locate and rate road hazards
- Merge repeat detections into one event
- Shared hazard map with filters + clustering
- Authority repair lifecycle + admin zones/teams

Non-functional

- Responsive uploads and map interaction
- Reliable asynchronous processing
- Secure, role-aware access control
- Accurate GPS mapping + 5 m de-duplication
- Clear status visible to every stakeholder

Precision 91.9% target ≥ 80%	Recall 85.2% target ≥ 70%	Processing Ratio 0.25–0.67× target ≤ 1.0× video length	Deduplication 5:1–8:1 target ≥ 3:1	E2E Latency 20–45 s target ≤ 5 min	Map Response < 300 ms target ≤ 2 s	GeoValid ≥ 0.99 target ≥ 0.95
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Conclusions

- RoadSenseAI delivers the full loop: capture -> detect -> map -> assign -> repair.
- The system combines AI detection with a practical city maintenance workflow.
- One pothole becomes one shared, evidence-based and managed repair story.

Future Work

- Dedicated mobile capture app for recording, GPS tracking and upload in one flow.
- Cloud deployment with scalable workers, managed MongoDB and production security hardening.
- Local labeled benchmark and continuous learning from authority decisions.